VARUN NATARAJ

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PROFESSIONAL SUMMARY

Software engineer and technical team leader with hands-on experience building end-to-end web platforms, PostgreSQL-backed services, and parallelized data-processing workflows. Led a 10-member team delivering public-sector election systems serving 1,000+ daily active and peak users, and a voter-ID generation platform associated with ₹15 crore+ in revenue. Skilled in Python backend engineering, React/TypeScript, full-stack delivery, Linux, Docker, and cloud-oriented development.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, HTML
Backend & APIs: Django, Flask, FastAPI, REST API development, Node.js
Frontend: React.js, TypeScript
Data & Infrastructure: PostgreSQL, Docker, Linux, AWS, Git, CI/CD
Engineering Practices: Full-stack development, database design, parallel processing, software architecture, Agile/Scrum, technical leadership

PROFESSIONAL EXPERIENCE

Gundal Business Private Limited
Sep 2025 – Present

Senior Software Engineer / Team Leader | Bengaluru, India

- Led a 10-member team in developing and delivering the Electoral-roll Matrix Application for Greater Bengaluru Authority (GBA) local-body election operations across three election phases generating ₹3 crore+ in revenue.
- Engineered an end-to-end full-stack election (matrix data collection, electoral-roll generation) management application using Python web frameworks, React/TypeScript, and PostgreSQL, supporting 1,000+ daily active and peak users.
- Architected parallelized voter-ID card generation software for approximately 90 lakh (9 million) voters across five city corporations; the application generated over ₹15 crore in revenue.
- Built and maintained client relationships with the State Election Commission and Greater Bengaluru Authority, aligning engineering delivery with stakeholder requirements.
- Contributed to a pilot for polling-station and assembly-part rationalization, supporting election-data organization and operational planning.
- Owned development across database design, backend services, frontend functionality, and deployment/DevOps workflows while coordinating engineering execution.

Key projects

- **Matrix Automation (Matrix v1):** Built the first version of the Matrix application in Flask, SQLAlchemy, and PostgreSQL, covering polling-station and section matrix entry, re-serialization, ward/PS mapping, TOTP two-factor authentication, CSRF protection, per-user feature controls, and ward-, zone-, and AC-wise progress reports with Excel and PDF exports.
- **Matrix App v2.0:** Re-architected Matrix into a multi-tenant platform with a FastAPI backend and React/TypeScript frontend, state-owned PostgreSQL databases per tenant (Karnataka, Maharashtra) with Alembic migrations, JWT-based authentication, and MinIO object storage, shipped as a Docker Compose stack behind an nginx reverse proxy with OpenAPI-documented REST APIs.
- **SVIC (Voter ID Cards):** Built the production pipeline behind voter-ID card generation, rendering bilingual Kannada card PDFs through assembly-constituency jobs split into polling-station units across CPU-parallel workers, with PostgreSQL-tracked job status and failure review, full-AC retry, batched reprints by voter ID, and MS SQL query/index tuning.
- **Marvel:** Built a bilingual (Kannada/English) address-resolution system for 4.8 lakh+ voter records across 390 polling parts in Assembly Constituency 175, grouping addresses into a locality → sublocality → building hierarchy with persistent place identities, atomic PostgreSQL run publication, gold-set evaluation, and a review browser that surfaces places split across polling parts.

HabbaOnline
Aug 2020 – Jul 2021

Co-founder / Software Developer | Python, Flask, SQL, Docker, AWS, Git

- Built an end-to-end digital platform connecting Indian festival workers with opportunities during COVID-19 lockdowns, developing application functionality across backend and deployment layers.
- Coordinated a small team and supported initial user acquisition, securing 30 registrations during festival campaigns.

Honeywell Technology Solutions
Jan 2018 – May 2018

Software Engineering Intern | Python, Bash, Git, Linux, PyTorch

- Developed and simulated multi-sensor data fusion for micro aerial vehicle pose estimation within the aerospace division.
- Implemented a neural-network prediction algorithm in PyTorch that achieved 92% accuracy in the project evaluation, using a multivariate Kalman-filter-based architecture for position estimation.
- Facilitated Scrum sessions, helping resolve team blockers and improve on-time delivery; reported a 20% team-efficiency improvement.

EDUCATION

BMS College of Engineering | B.E., Information Science | 2014–2018 | GPA: 8.87